

Week 1 — Lab Handout (Student)

Team Setup, Project Selection & Repository Scaffolding

Time: ~90 min in-class **Work as:** Your team (5 members) **AI use this week:** Allowed for brainstorming only. Log anything you use. **What you leave with:** A formed team, a chosen project, a protected Git repo, and a live task board.

Why this lab matters

You are not writing application code today. You are building the scaffolding your team will use for the next 12 weeks. Every later lab assumes this is in place and working. Teams that rush this step pay for it in Week 3.

Task 1 — Form your team and assign starter roles (15 min)

Form a team of **5**. Roles rotate over the term, but pick a starting owner for each:

- **Repo Lead** — owns branch protection settings and merge hygiene.
- **Board Lead** — owns the task board, keeps tickets current.
- **Scribe** — records decisions and the retrospective notes.
- **Two Builders** — everyone builds; these two have no extra admin role this sprint.

Roles are about *accountability*, not privilege. Everyone commits code. Everyone gets reviewed.

Deliverable checkpoint: team roster with names + starting roles, posted to your board.

Task 2 — Choose a local business problem (20 min)

Pick a real, local, small business or organization problem your app will solve. Good projects are **small, real, and CRUD-shaped**.

Your problem **must** satisfy all of these:

- ☐ Involves managing records users can **Create, Read, Update, and Delete**.
- ☐ Has at least **2-3 related record types** (e.g., a bakery's *orders, products, customers*).
- ☐ Is understandable to an outsider in one sentence.
- ☐ Is small enough to finish in 12 weeks — **not** "a full marketplace / social network."

Good examples: a barbershop appointment tracker, a sari-sari store inventory + sales log, a community library lending system, a tutoring-center session scheduler.

AI is allowed here: brainstorm 5 candidate problems with an AI assistant, then choose one *yourselves*. Save the prompt and the ideas — this goes in your Week 1 AI note.

Deliverable checkpoint: a one-paragraph problem statement + the 3-4 record types your app will manage.

Task 3 — Initialize a protected Git repository (30 min)

This course standardizes on GitHub. Everyone should already have a free account (see the Tooling Setup Guide). If your instructor is using **GitHub Classroom**, accept the assignment link first — it creates your team repo for you, and you can skip step 1.

1. **Repo Lead** creates one repository for the team (or accept the Classroom link). Add all 5 members as collaborators (**Settings → Collaborators**).
2. Everyone clones it locally and confirms they can pull.
3. Add a `README.md` with the project name, one-line description, and team roster. Commit it via a **branch + pull request**, not a direct push — practice the loop immediately.
4. Add a `.gitignore` appropriate to your (future) stack.
5. **Turn on branch protection for `main`** (**Settings → Branches → Add branch ruleset**, or **Add rule** for `main`): - [] Require a pull request before merging. - [] Require at least **1 approving review**. - [] Block direct pushes to `main` (do not allow bypass).
6. Test the protection: try to push a commit straight to `main`. It should be **rejected**. If it succeeds, protection is misconfigured — fix it.

Prefer GitLab or Bitbucket and have a reason? Clear it with your instructor first — the whole class supporting one platform is what keeps help fast. The branch-protection and pull-request steps are equivalent on each.

The Git collaboration loop you'll repeat all term: `pull main → branch → commit → push branch → open PR → get a review → merge`

Deliverable checkpoint: repo URL, screenshot of branch-protection settings, and one merged PR (the README).

Task 4 — Stand up your task board (15 min)

Use **GitHub Projects** (recommended — it lives next to your code, so a ticket can link to the pull request that closes it, which makes your individual grade easy to verify). Create it from your repo's **Projects** tab → **New project** → **Board** template. If your instructor prefers a simpler standalone board, **Trello** works too — one board per team.

Set up at minimum these columns: Backlog → To Do → In Progress → In Review → Done

Then:

- ☐ Create a ticket for each thing you did today (team roster, problem statement, repo setup, board setup).
- ☐ **Assign an owner** to every ticket. A ticket with no owner does not count.
- ☐ Move completed tickets to **Done**.

The board is the source of truth for your individual grade. If your name isn't on tickets and your commits don't exist, there is no evidence you contributed.

Deliverable checkpoint: board URL with at least 4 owned, correctly-placed tickets.

End-of-Lab Checklist (submit to your instructor)

Post this list to your board or hand it in:

- ☐ Team roster with starting roles
 - ☐ One-paragraph problem statement + 3–4 record types
 - ☐ Git repo URL with `main` branch protection **enabled and tested**
 - ☐ At least one merged pull request (the README)
 - ☐ Task board URL with 4+ owned tickets
 - ☐ Week 1 AI note: the brainstorming prompt(s) you used and the ideas they produced
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Looking ahead to Week 2

Next week you turn this problem into a **feature backlog covering full CRUD** and draw **wireframes** for each screen. That work, plus this scaffolding, becomes **Deliverable 1 (Scoping & Wireframes, 15%)**, due end of Week 2. Read the Deliverable 1 spec before next session.